



STUDENT LAB ACTIVITY REPORT

Course Name: Introduction to Machine Learning

Lab Teacher: Dr. Aymen TRIGUI

Course Code : 462 CCS-3

Student ID: 445801958

Section Number : S.No

Student Name: Abdulmalik Hussein Al-Haif

LAB ACTIVITY - 3

(1.5 Marks)

In continuation of Activity 2 Perform the following operations

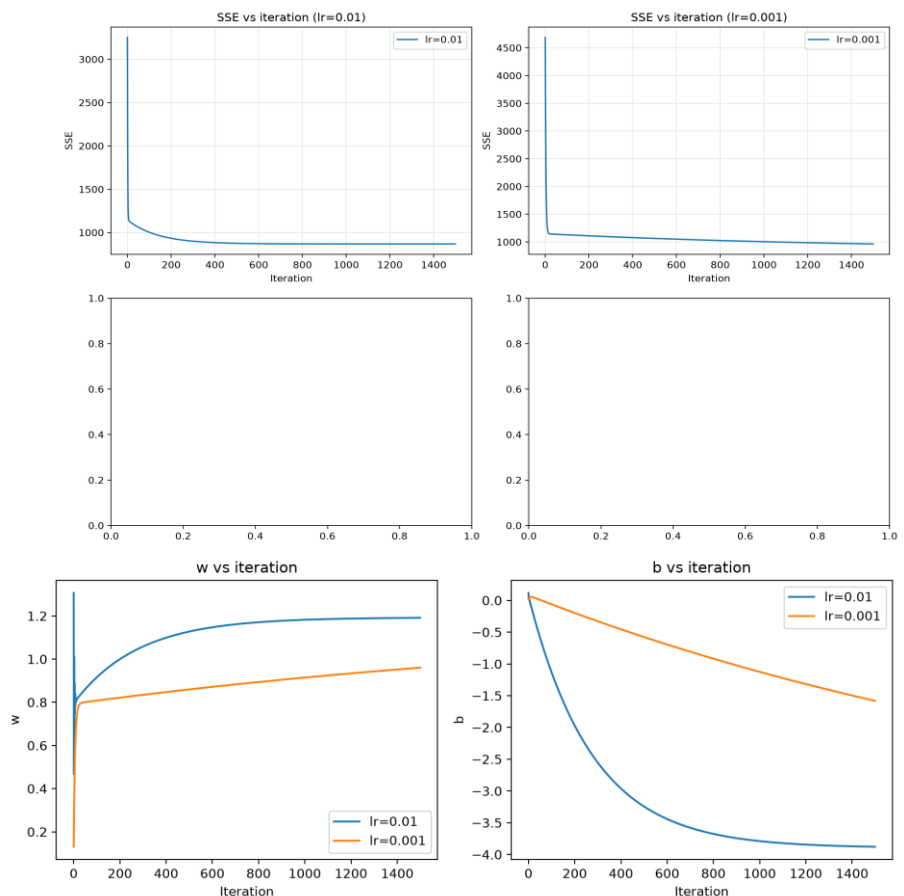
1. Implement the GD algorithm to compute the linear regression model parameters that fit your dataset.
2. Plot the Graph: SSE Versus Iteration, w Versus Iteration, b Versus Iteration
3. Plot the cost $J(w, b)$ versus the learning rate (take two different values of learning rate)
4. Evaluate the obtained model's performance concerning the learning rate's chosen values.

Gradient descent results:

- Best learning rate: 0.0100
- Final w : 1.191261
- Final b : -3.878138
- Final SSE: 868.537945

Other learning rates:

- $lr=0.0100$: $w=1.191261$, $b=-3.878138$, $SSE=868.537945$
- $lr=0.0010$: $w=0.960498$, $b=-1.581088$, $SSE=963.160110$



Marks Obtained	/1.5
Teacher's Signature:	